



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

PHASES OF COMPILER AND THEIR ROLE

TECHNICAL PRESENTATION

CSA1402-COMPILER DESIGN

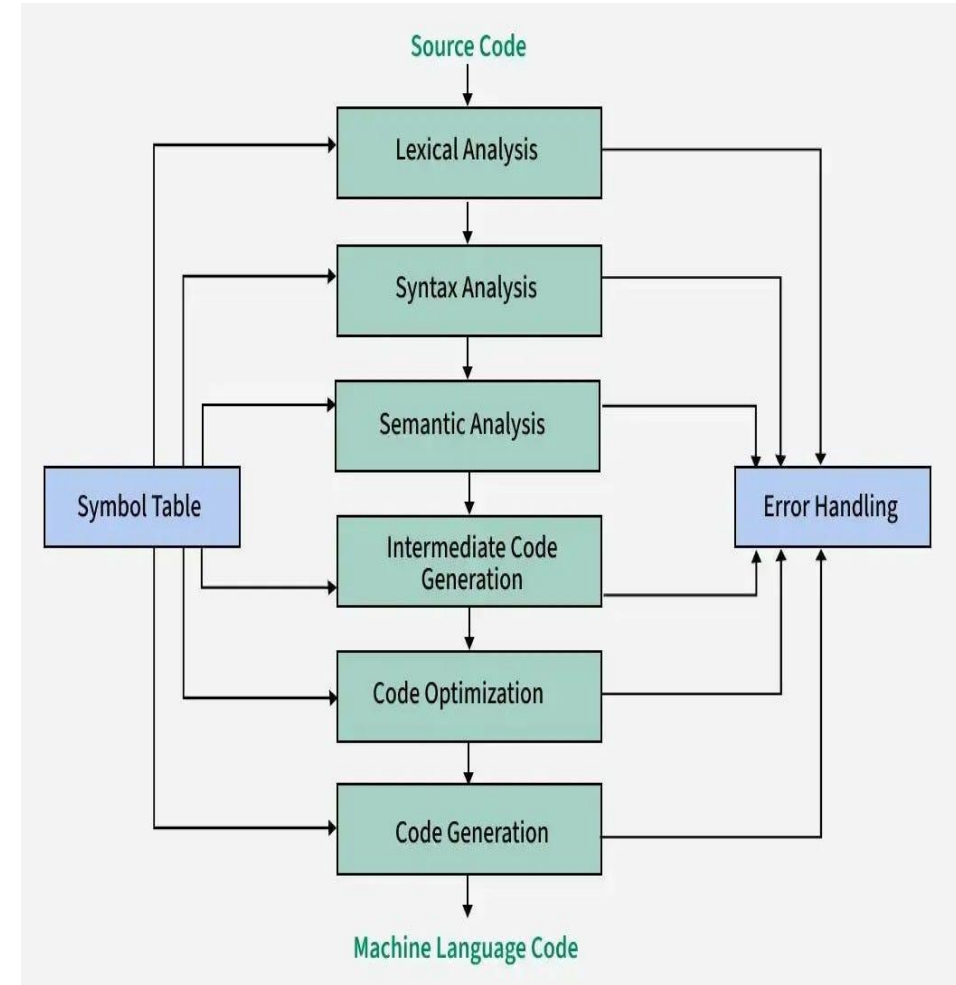
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Phases of compiler

Six Main Phases:

- Lexical Analysis
- Syntax Analysis
- Semantic Analysis
- Intermediate Code Generation
- Code Optimization
- Code Generation



Lexical Analysis

- **Role**
- First phase of compiler
- Reads source code character by character
- Groups characters into tokens
- Removes spaces and comments
- Detects lexical errors

Example:

```
int count = 5;
```

Syntax Analysis

Role

- Also called Parsing
- Checks grammar rules
- Creates Parse Tree
- Detects syntax errors

Example:

```
sum = a + b;
```

Semantic Analysis

Role

- Checks meaning of the program
- Verifies variable declaration
- Performs type checking
- Detects semantic errors

Example: `int x;`

`x = "Hello";`

Intermediate Code Generation

Role

- Converts program into intermediate representation
- Makes compiler machine independent
- Used before optimization

Example: $a = b + c$

Intermediate Code:

$t1 = b + c$

$a = t1$

Code Optimization & Code Generation

Code Optimization

- Removes unnecessary instructions
- Reduces execution time
- Reduces memory usage

Eg: $x = y * 1$

Optimized

$x = y$

Code Generation

- Final phase
- Produces assembly or machine code
- Allocates CPU registers
- Generates executable program

Eg: LOAD R1, a

ADD R1, b

STORE R1, c

Code:

```
techpresentation.cpp X
1  #include <stdio.h>      // Header file
2
3  int main()              // Main function
4  {
5      // Semantic Analysis:
6      // Variables are declared with correct data types
7      int a = 10;
8      int b = 20;
9      int sum;
10
11     // Intermediate Code Generation:
12     // Compiler converts this statement into intermediate code
13     sum = a + b;
14
15     // Code Optimization:
16     // Compiler removes unnecessary operations (if any)
17     // and makes the code efficient
18
19     // Code Generation:
20     // Compiler converts the optimized code into machine code
21     printf("Sum = %d", sum);
22
23     return 0;
24 }
```

Output:

```
C:\Users\pooja\OneDrive\Des X + - □ X
Sum = 30
-----
Process exited after 0.1747 seconds with return value 0
Press any key to continue . . . |
```



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Conclusion

- Compiler translates high-level language into machine code.
- Each phase performs a specific task.
- Symbol table and error handling support all phases.
- Efficient compilation results in optimized executable programs.

THANK YOU



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